Grasp-Conditioned Planning and Learning for Multi-Robot Transport

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***Abstract*— Cooperative transport couples three feasibility conditions: robots must reach compatible grasps, the resulting assembly must clear the environment, and local control must maintain support while following the object reference. We develop a grasp-conditioned planning and learning framework for two wheeled bimanual robots. Reachable carrier placements define an assembly footprint and aperture-constrained search; object-level MPC and a shared acquisition supervisor coordinate identical local actors. Fixed object-to-hand transforms extend the reference to spatial reorientation and native-contact lift control. Across nine scenes, a geometric screen retains 45,917 of 2,110,211 sampled opposed layouts. Under a matched search budget, aperture sampling verifies 16/16 routes versus 5/16 for uniform sampling. Frozen welded-retention cohorts complete 35/40 and, with a different actor and grasp configuration, 46/48 missions. Spatial probes expose substantial internal weld reactions despite accurate object tracking. Calibrating the native grasp reference and training a staged local policy yields 200/200 near-grasp lift-and-hold trials at the nominal randomization level with nearly stationary bases. These results connect geometric compatibility, coordinated progression and contact validity while preserving the distinct scope of each evaluation.**

I. Introduction

Two robots carrying furniture through a doorway must solve more than an object-path problem. Their bases may not fit where the payload fits; their hands may reach a bar without establishing a usable grasp; and a feasible path may become incompatible with the formation attained during acquisition. These dependencies make grasp selection, motion planning and local control parts of the same task, even when each component is implemented separately.

Grasp-aware cooperative planners couple contact selection with navigation [1], [2], while learned transport policies address the response of physically coupled robots [3], [4]. Their integration requires a consistent interface: which grasp geometry is assumed, which common motion is requested, and what constitutes successful execution? Without this interface, an acquisition controller can produce a formation different from the one planned, or a tracking reward can favor hand poses that conflict with the contact geometry.

We formulate cooperative transport around a shared object reference and fixed object-to-hand transforms. At the planning level, a standing-band screen and whole-body IK identify candidate carrier placements. The attained post-lift layout defines a payload-plus-bases footprint whose aperture constraints guide global search and local MPC. At execution, identical actors use robot-relative observations and a common motion reference; a supervisor coordinates asynchronous attachment and lift progression. The object-to-hand mapping also supplies spatial roll commands and the pose targets used by a separate native-contact learning stage.

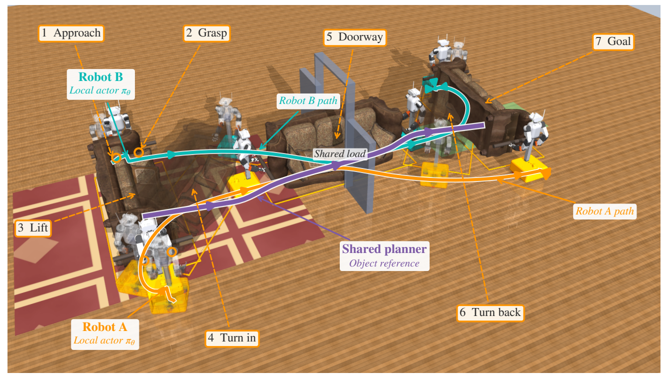

Fig. 1. Cooperative acquisition, lift and doorway transport. The object reference must accommodate the carriers and preserve their grasp assignments through changes in motion and support.

The evaluation follows this hierarchy. Geometric experiments measure the available grasp space and the effect of aperture sampling. Frozen missions measure acquisition through goal arrival. Spatial and native-contact studies then examine what geometric path completion leaves unresolved: grasp loading, reference attainability and sustained physical support. This organization connects the project's components through testable interfaces rather than pooling results from different controllers into one success rate.

The contributions are:

- A grasp-conditioned geometric formulation connecting reachable carrier placements, aperture sampling and a bound on clearance loss under formation drift.
- A shared-reference architecture for asynchronous acquisition, planar transport and fixed-grasp spatial commands, with an explicit account of the information supplied to each local actor.
- A multilevel simulation evaluation, including a matched search study, frozen mission cohorts, spatial load diagnostics and calibrated native-contact lift learning with task acceptance separated from reward shaping.

The complete transport cohorts use welded retention. Native-contact results cover near-grasp lift and hold, not frictional-retention doorway transport. This distinction defines the physical scope of the current framework.

II. Related Work

a) Cooperative planning: Paired-grasp planning couples contact selection to object transport [1]; semi-coupled methods combine an object path with local redundancy resolution [2]. Distributed MPC coordinates coupled motion through exchanged predictions [5]. Our formulation fixes the

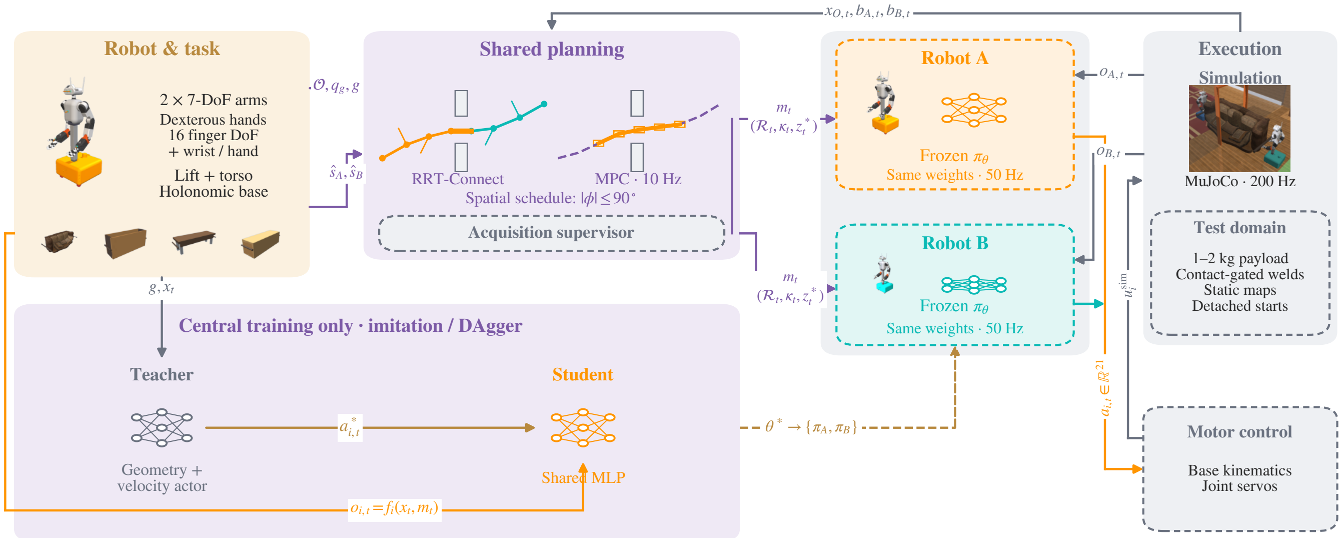

Fig. 2. The evaluated full-mission architecture. Acquired carrier geometry conditions shared planning; phase and height references coordinate local actors trained by centralized imitation. The native-contact study uses the same object-to-hand reference principle with a separate 23-action PPO actor, explicit finger-contact measurements and no grasp welds.

grasp assignments and conditions planning on the carrier placement attained after lift. The aperture construction and drift bound make the resulting reduced collision model explicit.

b) Learned physical coordination: Shared policies can support multiple embodiments [6] and concurrent collaboration [7], but their coordination properties also depend on observations and task structure. In cooperative carrying, decMBC conditions local control on rigid attachment geometry [3], whereas decPLM uses an arm/base hierarchy and contact-constellation rewards to maintain surface contact [4]. Sequential Asymmetric Imitation varies partner behavior during training to develop waiting and yielding behaviors [8]. An et al. use a dynamic reward curriculum to learn approach, pickup, transport and release with one-armed legged robots [9]. Our full-mission study uses supervised readiness and welded retention; the separate native-contact stage adapts curriculum shaping to four-hand pose acceptance. Its restricted lift task is not an equivalent evaluation of end-to-end pick-and-place.

c) Object-centric dexterous reinforcement learning: SimToolReal learns tool manipulation with object-pose goals and procedural geometry [10]. Play2Perfect pretrains manipulation through play, then adapts the policy to assembly with sparse-reward RL [11]. These single-arm results motivate separating contact acquisition from downstream task refinement. They do not establish four-hand load sharing, asynchronous acquisition, or grasp-loss recovery for our mobile team. Our contact-reference calibration and near-grasp lift study retain a separate protocol from the welded transport cohorts.

d) Interaction feedback and shared references: Physical interaction can communicate intent through partner actions and haptic feedback [12], [13]. PAINT estimates an interaction wrench from proprioceptive history to guide locomotion [14]. Our actors instead observe object state and track an explicit shared reference. The distinction is consequential: object motion and readiness enter through a common planning interface, rather than being inferred from a partner's force. We specify this information flow to delimit the coordination capability measured by the experiments.

III. Grasp-Aware Planning

The task specifies an object, assigned hand–rail contacts, a static map and a goal pose. Two wheeled bimanual robots acquire and lift the object before planning its transport. The attained carrier layout defines a planar assembly model for global search; object-level MPC supplies a common motion reference to the local actors (Fig. 2).

A. Grasp geometry as a team constraint

Let $q = (t, \theta) \in \mathrm{SE}(2)$, $t = (x, y)$, denote object pose and $s_i = (u_i, v_i)$ the nominal position of base i in the object frame. With payload rectangle $\mathcal{P}$ of dimensions (L, D) and base-disc radii r_i,

$$\mathcal{C}(s) = \mathcal{P} \cup \bigcup_{i=1}^{2} B(s_i, r_i), \qquad \mathcal{C}_q = t + R(\theta)\mathcal{C}(s). \tag{1}$$

Calibrated object profiles specify reachable grasps. Base offsets s_i are measured after lift and held fixed for route planning. This is a formation approximation: articulated arms allow the physical bases to move relative to the object. For obstacles $\mathcal{O}$ and signed clearance d, the configuration-space problem [15] is

$$\begin{aligned} &\gamma : [0,1] \to \mathrm{SE}(2), \quad \gamma(0) = q_{\mathrm{lift}}, \quad \gamma(1) = q_g, \\ &d\big(\mathcal{C}_{\gamma(\tau)}, \mathcal{O}\big) \geq \varepsilon \quad \text{for all } \tau \in [0,1]. \end{aligned} \tag{2}$$

The collision model excludes supports below a specified carrying-height bound and represents the payload and bases, not the articulated swept volume.

a) Admissible carrier placements: Before executing a fixed grasp assignment, candidate base stations are screened in the object frame. For usable horizontal reach R_i at the carrying height, a necessary planar condition is

$$r_i \leq d(s_i, \mathcal{P}) \leq R_i, \qquad \|s_1 - s_2\| \geq r_1 + r_2. \tag{3}$$

For a rectangle, $d((u,v),\mathcal{P}) = \|((|u| - L/2)_+, (|v| - D/2)_+)\|_2$. This standing band excludes base overlap with the payload and unreachable standoffs; it is a candidate screen, not a sufficient arm-pose test. Opposed placements are additionally filtered by aperture feasibility and checked with whole-body IK. Unequal carrier offsets are permitted, so geometric symmetry is not imposed on the team.

The sensitivity to formation drift can be stated without assuming rigid attachment of the bases. Let s^0 denote planning offsets and $\delta_s(t) = \max_i \|s_i(t) - s_i^0\|$. At the same object pose q, translation of each base disc by at most δ_s gives

$$\operatorname{dist}(\mathcal{C}_q(s(t)),\mathcal{O}) \geq \operatorname{dist}(\mathcal{C}_q(s^0),\mathcal{O}) - \delta_s(t). \tag{4}$$

To see this, pair every point in each translated disc with the point at the same relative position in its planning disc. Paired points differ by at most δ_s, and payload points are unchanged. Distance to the fixed obstacle set is 1-Lipschitz, yielding Eq. (4) after taking infima. The bound applies to unsigned set distance at the same object pose; strictly positive planned distance greater than δ_s preserves separation of this footprint. It excludes path error and articulated swept volume, and is evaluated retrospectively rather than enforced online.

For a centered opening of width W, define $w_{\mathcal{P}} = L|\sin\theta| + D|\cos\theta|$ and

$$\begin{aligned} w_i(\theta;s) &= 2\big(|u_i\sin\theta + v_i\cos\theta| + r_i\big),\\ \bar{w}(\theta;s) &= \max\{w_{\mathcal{P}}, w_1(\theta;s), w_2(\theta;s)\}. \end{aligned} \tag{5}$$

The aperture condition $\bar{w} \leq W - 2\varepsilon$ is exact for mirrored, equal-radius layouts and conservative otherwise. Yaw scanning and boundary bisection identify admissible intervals.

The corresponding gate-pose set couples orientation and lateral placement:

$$\mathcal{G} = \left\{(x_d, y, \theta) : |y - y_d| \leq \frac{W - \bar{w}(\theta;s)}{2} - \varepsilon\right\}. \tag{6}$$

Here (x_d, y_d) is the opening center; a negative bound excludes that yaw. This sampling constraint does not certify clearance elsewhere. For the recorded beam handoff, the object-only admissible yaw window is 25.8°; including the carriers reduces it to 5.5° (Fig. 5).

Gate-biased RRT-Connect [16] samples the opposite root, aperture set and workspace with probabilities $(0.10, 0.35, 0.55)$. Steering uses $d_q = \|t_a - t_b\|_2 + 0.6|\operatorname{wrap}(\theta_a - \theta_b)|$ and 0.35 m-equivalent extensions. Search permits three seeded attempts of 20,000 iterations, followed by shortcutting and clearance-checked smoothing. Swept collision checks use 5 mm increments and a 22 mm margin; an independent 10 mm-resampling verifier requires $\varepsilon = 20$ mm along the delivered path, including joins. These finite-resolution checks implement Eq. (2); failed verification rejects the route.

B. Shared receding-horizon object reference

Object MPC [17] predicts q_k under object-frame twist $\nu_k = (v_x, v_y, \omega)$ with timestep h:

$$q_{k+1} = f(q_k, \nu_k) = q_k + h\ \operatorname{diag}(R(\theta_k), 1)\nu_k. \tag{7}$$

At each solve, measured object pose and base offsets update the model. Offsets are parameters, not control decisions. Monotone measured path progress selects $\bar{q}_{1:N}$, so elapsed time alone cannot advance the reference during a stall. With $e_k = q_k - \bar{q}_k$ and locally unwrapped yaw, successive convexification solves

$$\begin{aligned} \min_{q_{1:N},\nu_{0:N-1},\sigma\geq 0}\ & \textstyle\sum_{k=1}^{N-1}\|e_k\|_Q^2 + \|e_N\|_{Q_f}^2 \\ & + \textstyle\sum_{k=0}^{N-1}\|\nu_k\|_{R_\nu}^2 + \lambda\|\sigma\|_1 \\ \text{s.t.}\quad & q_0 = \hat{q},\quad q_{k+1} = A_k q_k + B_k\nu_k + c_k, \\ & a_{jk}^\top q_k + \sigma_{jk} \geq b_{jk} + \varepsilon, \\ & |\nu_k| \leq \nu_{\max},\quad |\nu_k - \nu_{k-1}| \leq h\dot{\nu}_{\max}, \\ & \|\nu_{xy,k} + \omega_k J\hat{s}_i\|_\infty \leq 1.2\ \text{m/s}. \end{aligned} \tag{8}$$

Here J rotates by 90° and $\hat{s}_i$ is the measured base offset. Affine dynamics approximate Eq. (7); separating planes approximate obstacle clearance with individual nonnegative slacks σ_{jk}. The frozen profile uses $N = 20$, $h = 0.1$ s, $Q = \operatorname{diag}(12, 12, 30)$, $Q_f = 40Q$, $R_\nu = \operatorname{diag}(0.6, 0.6, 0.8)$ and $\lambda = 4000$. Two warm-started convexification passes use OSQP [18]. Limits are 0.12 m/s per translation axis and 0.10 rad/s yaw, with slew bounds of 0.15 m/s^2 and 0.10 rad/s^2. The first control is accepted only if the solve succeeds, the command is finite, and maximum slack is at most 10 mm; otherwise the bridge commands zero twist. Execution clearance is measured independently of the QP slacks.

For world-frame station offset ρ_i from the object center, the common reference induces rigid-motion feedforward

$$v_i^{\text{ff}} = R(\theta)(v_x, v_y)^\top + \omega\hat{z}\times\rho_i, \qquad \omega_i^{\text{ff}} = \omega. \tag{9}$$

Each robot transforms this reference into its base frame and adds bounded learned residuals. Both planners use the same assembly geometry and requested clearance. Reference updates preserve the continuous physical state.

C. Fixed-grasp object reorientation

Spatial pose tracking extends the reference interface; it does not replace planar RRT/MPC with a collision-free SE(3) search. Let $T_O = (R_O, p_O) \in \mathrm{SE}(3)$ and let $\bar{T}_{ih}$ be the fixed object-to-hand transform for robot i, hand $h \in \{L, R\}$. The reference satisfies

$$\begin{aligned} T_{ih}^*(t) &= T_O^*(t)\bar{T}_{ih}, \\ R_O^*(t) &= R_{O,0}\exp\big(\phi(t)[e_x]_\times\big). \end{aligned} \tag{10}$$

where $R_{O,0}$ is the lifted orientation and e_x the object's long axis. Upper/lower handles provide distinct hand targets during roll. We adopt the fixed-grasp physical operating envelope

$$|\phi(t)| \leq \phi_{\max} = \pi/2, \qquad \bar{T}_{ih}(t) = \bar{T}_{ih}(0). \tag{11}$$

Beyond 90°, continuing the same hand assignment twists the arms and requires switching hands along the handles. Such contact reassignment and regrasp planning are outside this paper. This mechanism-specific envelope is not a guarantee of reachability for every pose within it. It bounds object roll relative to the lifted frame, not world-frame yaw during transport. Together, the planar and spatial tests examine cooperative translation, yaw and out-of-plane rotation (Fig. 3); independent pitch-axis control is not evaluated.

The high-level reference specifies a clearance lift followed by roll. Over a segment beginning at t_a with duration τ, set $u = (t - t_a)/\tau \in [0, 1]$ and interpolate

$$\begin{aligned} s(u) &= 10u^3 - 15u^4 + 6u^5, \\ p^*(u) &= p_a + s(u)(p_b - p_a), \\ R^*(u) &= \exp\bigl(s(u)\log(R_b R_a^\top)\bigr)R_a. \end{aligned} \tag{12}$$

Differentiation supplies the centroid twist. Unlike planar MPC's measured path progress, these pose probes use a timed schedule after supervised lift. The local actor receives pose/twist errors and hand targets in its base frame; the geometric teacher instead maps hand errors to bounded Jacobian rates with joint-limit and selected self-clearance constraints. Neither controller chooses the requested endpoint. Large commanded roll replaces the planar upright-object guard with pose-tracking and contact guards; the 2° base-inclination limit remains.

IV. Local Control and Coordination

The simulated team consists of two holonomic mobile bases, each carrying a two-stage lift, a pitching torso and two 7-DoF arms. The planar policy commands base motion and 18 upper-body coordinates; a stage-dependent closure rule actuates the hands. Payload mass is 1–2 kg. The supplement describes the prospective hardware interface separately.

A. Shared policy and local observations

In the frozen planar cohort, each robot executes $a_i = \pi_\theta(o_i)$ at 50 Hz with identical weights; MuJoCo dynamics run at 200 Hz [19]. The 157-dimensional observation includes joint state and tracking errors, base motion and gravity, object-relative pose/twist, grasp geometry, palm-target and docking errors, phase/height references, previous action and commanded motion. It excludes partner joint/velocity streams. Simulation supplies object state and six clipped constraint-multiplier channels for grasp feedback.

For base position p_i and yaw ψ_i, relative features are

$$\begin{aligned} \tilde{p}_i &= R(-\psi_i)(t - p_i), \qquad \tilde{s}_i = R(\theta - \psi_i)s_i, \\ c_i &= \bigl(\cos 2(\theta - \psi_i),\ \sin 2(\theta - \psi_i)\bigr). \end{aligned} \tag{13}$$

The double-angle feature represents footprint symmetry; assigned hand targets retain contact identity. This common coordinate convention permits weight sharing between opposed carriers without assuming policy equivariance. The 512–256–128 ELU MLP produces three base and 18 upper-body actions. Arm targets are posture residuals; lift actions specify a bounded 0.06 m/s rate during lifting and position residuals around the attained carrying posture during transport.

B. Acquisition and asynchronous progression

Assigned rails define the approach targets. The geometric teacher stages, docks and refines wrist pose within 40 mm. Panel/beam profiles use 0.18 m hand spacing instead of 0.40 m and reserve 60 mm downward lift travel. Robot i attaches when both assigned palms are within 35 mm and 30° of their targets, base speed is below 0.10 m/s, and both palm/finger contacts apply more than 0.1 N compression to the correct rails. Weld constraints then retain the attained transforms [20]; subsequent frictional slip is not modeled.

One actor starts immediately and the other after $\Delta \in [0, 8]$ s; the delayed robot holds its initial command until activation. Let b_i indicate attachment and z_0 initial object height. The attached robot's height reference is

$$z_i^* = \begin{cases} z_0 + 0.04 \text{ m}, & b_i = 1,\ b_1 b_2 = 0, \\ z_0 + 0.12 \text{ m}, & b_1 b_2 = 1. \end{cases} \tag{14}$$

During early lift, the base holds and upward action is clipped at a 40 mm centroid rise. Remaining support permits this partial lift while the late robot tracks its moving grasp targets. Joint attachment releases the full lift. Transport begins after height error remains below 20 mm and object roll/pitch below 0.06 rad for 25 control steps (0.5 s). The same actor then tracks MPC references without resetting the physical state. Gated-start comparisons instead synchronize controller availability and attachment.

C. Information and physical assumptions

Let x_t be the simulator state, $b_{i,t}$ attachment status, and $m_t = (\mathcal{R}_t, \kappa_t, z_t^*)$ the shared reference: motion, phase and height. The evaluated information flow is

$$\begin{aligned} m_t &= \mathcal{H}(x_{O,t}, b_{1,t}, b_{2,t}, g), \\ o_{i,t} &= f_i(x_t, m_t), \qquad a_{i,t} = \pi_\theta(o_{i,t}), \\ \mathcal{D} &= \bigcup_{i,t}\{(o_{i,t}, a_{i,t}^*)\}, \\ \theta^* &= \arg\min_\theta \sum_{(o,a^*)\in\mathcal{D}} \|\pi_\theta(o) - a^*\|_W^2. \end{aligned} \tag{15}$$

Here g is task geometry, $\mathcal{R}_t$ the motion reference and W a diagonal action-weight matrix. Central training pools teacher-labeled samples from both robots; teacher labels may use simulator geometry and supervised progression. Student execution retains identical weights and normalization, but no teacher labels, optimizer state or partner joint/velocity streams.

Shared memory delivers simulator object state, 10 Hz planar MPC references and supervisor-selected phase/height to the 50 Hz actors. Spatial probes sample the common pose schedule at the control rate. These references convey joint readiness even without direct actor-to-actor messages. Configured update rates are not measured network traffic.

The frozen planar actor's six clipped feedback channels aggregate equality constraint multipliers, not calibrated wrist wrenches. Welds can sustain tensile forces and moments without frictional retention limits. The supplement defines the feature scaling and physical assumptions; no load-capacity or actuator-saturation claim is made.

V. Contact-Feasible Local Learning

Welded retention makes the planning/control interface testable but removes frictional grasp failure. We therefore instantiate the same object-to-hand reference in a separate native-contact task. The policy controls two scalar gripper closures in addition to the 21 base/upper-body channels. Two 128-unit ELU layers map eight 352-dimensional local history frames to 23 actions. Weights are shared across robots; this is a different actor from the 157-input planar policy. Native MuJoCo contacts supply the holding forces.

A. Calibrating an attainable reference

The transform $\bar{T}_{ih}$ in Eq. (10) must describe a pose that the closed hand can occupy. Parameterize its position by palm height b below the bar and forward displacement f. The earlier $(-40, +45)$ mm assignment produces 19.1–20.4 mm pinky/rail overlap when a measured closed hand is placed at that target. Its successful physical grip lies below the target, so pose alignment and contact retention conflict.

We screen (b, f) by scripted closure and lift, measuring penetration and simultaneous four-hand qualification. A refined sweep selects $(-65, +70)$ mm at zero approach standoff. Across three seeds and three masses (0.2, 0.3 and 0.4 kg), this candidate sustains 296–306 qualified control steps in a 350-step probe. Moving the base back is not necessarily safer: reset IK preserves the palm target while changing which finger segments approach the rail. The reference and reset posture must therefore be validated together.

B. Task acceptance and curriculum shaping

Let c_{ih} denote a qualified native grasp: contact opposition at least 0.7, summed normal force at least 2 N and force-weighted slip at most 0.04 m/s. These measured proxies do not certify force closure. For object errors (e_p, e_R) and palm errors (d_{ih}, α_{ih}) to $T^*_O \bar{T}_{ih}$, acceptance requires a continuous 1 s interval satisfying

$$\begin{aligned} &\|e_p\| < 0.04\,\mathrm{m}, \quad \|e_R\| < 0.08\,\mathrm{rad}, \\ &d_{ih} < 0.04\,\mathrm{m}, \quad \alpha_{ih} < 0.175\,\mathrm{rad}, \quad c_{ih} = 1\ \forall i, h, \\ &\|v_O\| < 0.05\,\mathrm{m/s}, \quad \|\omega_O\| < 0.2\,\mathrm{rad/s}, \end{aligned} \tag{16}$$

with no physical failure. Both object and hand errors matter: reaching the object goal alone does not establish retained grasps. A separate graded score reports partial progress without relaxing this conjunction.

PPO [21] optimizes contact and pose progress with dynamic shaping inspired by [9]. Terms belong to approach, contact, grasped, lifted, tracking or held stages. After each episode, a scale a_j multiplies the stored base weight of shaping term j:

$$a_j^+ = \begin{cases} \min(1.02 a_j, 1), & j \text{ belongs to an active stage,} \\ \max(a_j/1.02, 0.05), & \text{otherwise.} \end{cases} \tag{17}$$

The highest stage reached by at least half of the recent window (up to 20 episodes), and its next stage, are active. Terminal success/failure, safety and regularization terms retain their weights. Thus shaping changes with progress while the acceptance test and terminal task remain fixed.

The initial lift stage uses open hands near the calibrated reference, 0.3 kg nominal mass, high sliding friction (3–4), and a 1 N m hand motor torque limit. Translation is capped at 1 mm/s, so the task isolates closure, a 6–14 cm commanded lift and hold. Gripper output biases favor closing at initialization but remain trainable. Eight environments collect 128-step rollouts; PPO uses four epochs, clipping 0.2 and initial learning rate 10^{-4}. The evaluated checkpoint contains 81,920 environment steps. Full reward weights, initialization and reset-domain provenance are supplied in the supplement.

TABLE I

EVALUATION SCOPE. EACH ROW HAS ITS OWN PROTOCOL.

Study	Task and retention	Outcome
Grasp screen	Planar geometry, 9 scenes	45,917 layouts
Matched search	One scene; 16 seeds/method	16/16 vs. 5/16
Mission I	Detached; welded; 10 scenes	35/40
Mission II	Detached; welded; 12 scenes	46/48
Spatial probes	Welded; geometric teacher	45°, 90°
Native lift	Near-grasp; contact; nominal	200/200

TABLE II

SCENE GEOMETRY AND FROZEN OUTCOMES: FOUR TRIALS PER SCENE.

Scene	**Object**	$L \times D \times H$ **(m)**	S (%)	T_L	T_G
Living	Sofa	$2.73 \times 1.12 \times 0.93$	75	9.0	98.3
Bedroom	Wardrobe	$2.10 \times 0.62 \times 0.95$	100	11.7	90.9
Dining	Table	$2.40 \times 0.76 \times 0.75$	100	8.6	94.9
Dock	Case	$2.80 \times 0.82 \times 0.80$	100	9.3	106.3
Warehouse	Crate	$2.20 \times 0.70 \times 0.65$	100	9.7	96.4
Gallery	Panel	$1.90 \times 0.09 \times 1.25$	50	11.6	93.1
Workshop	Beam	$3.20 \times 0.20 \times 0.20$	75	15.0	122.1
Office	Table	$2.40 \times 1.10 \times 0.74$	75	9.1	93.5
Apartment	Sofa	$2.10 \times 0.86 \times 0.78$	100	10.7	91.8
Arena	Sofa	$2.10 \times 0.90 \times 0.83$	100	11.3	40.3
All (40)		–	87.5	10.6	95.4

Nominal dimensions; variation ±5%. T_L/T_G: median seconds to completed lift/first goal among passes. Simulation time; planning compute is excluded.

VI. Experimental Evaluation

We evaluate four linked questions: which grasp layouts admit passage; whether the aperture prior improves search under a fixed budget; whether the shared-reference architecture completes acquisition-to-goal missions; and which physical constraints remain after geometric tracking succeeds. Table I separates the protocols and their denominators.

A. Training and evaluation protocol

The planar actor is trained by imitation and DAgger [22]. A geometric teacher labels acquisition/lift; a saved velocity actor labels transport. Three 30-episode aggregation rounds produce a 524,490-sample dataset, including 345,484 imported development samples. Training minimizes phase-balanced weighted action MSE across both robots; no subsequent PPO update is applied. The supplement specifies data provenance, normalization and optimization settings.

The frozen test uses four prespecified held-out seeds in each of ten collection scene families: nine doorway environments and an open arena (Table II). Each mission starts detached and has a 180 s simulation budget. Randomization covers mass $U[1, 2]$ kg, dimension factors $U[0.95, 1.05]$, rail friction $U[0.9, 1.8]$, base offsets 0.08–0.50 m, headings ±20° and activation delays 0–8 s. Object XY and yaw additionally vary by ±0.10 m and ±0.20 rad. Only one trained actor is tested; unseen scene families and training-seed variation are not evaluated.

Physical termination F includes forbidden world contact, object height outside $[z_{\text{rest}} - 0.12, z_{\text{rest}} + 0.30]$ m, object

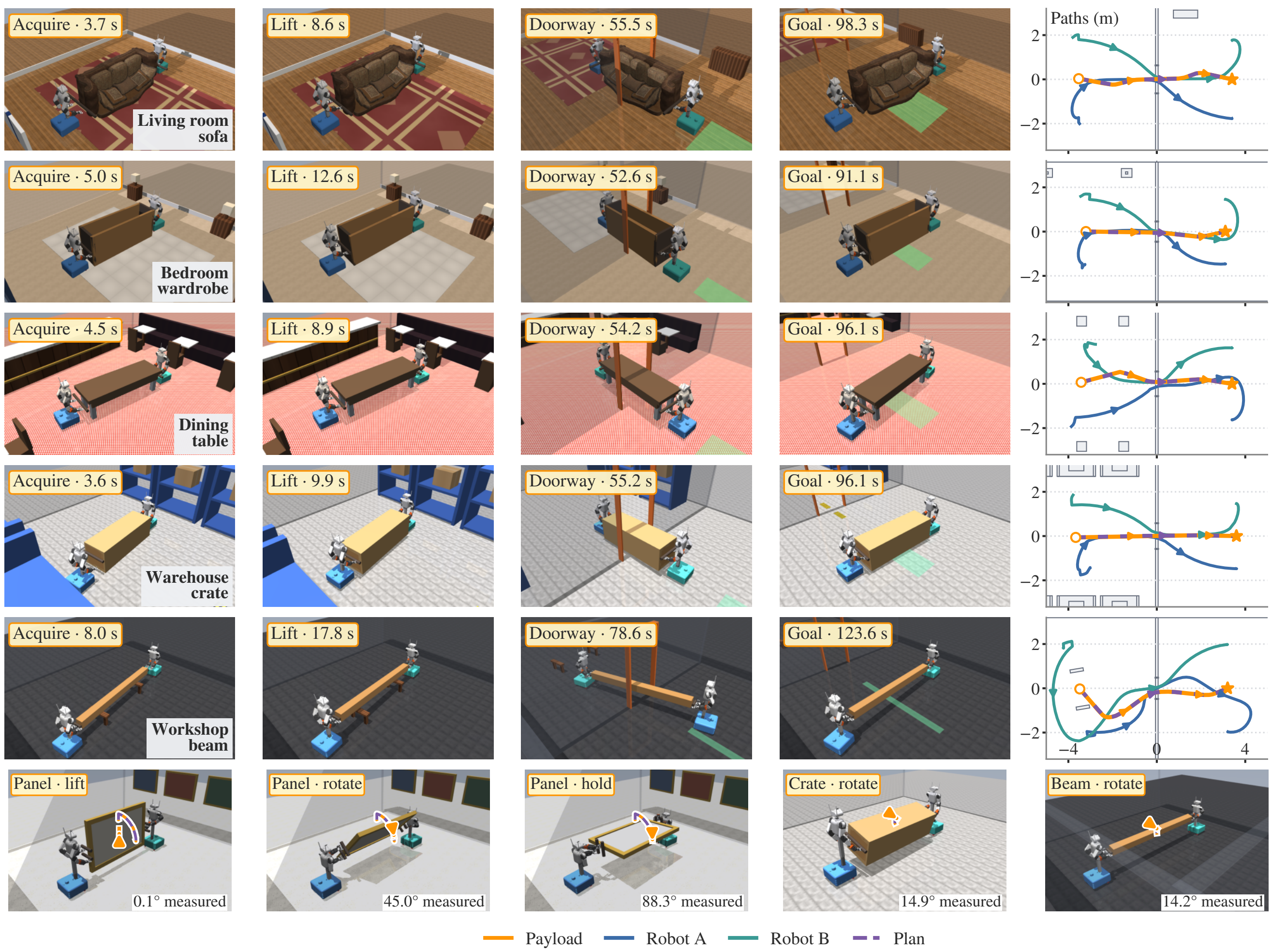

Fig. 3. Recorded acquisition, lift and transport in five scenes (top), and separate geometric-teacher roll probes (bottom). Reference and measured paths are overlaid. Spatial probes do not include doorway passage.

roll/pitch above 0.6 rad, or base inclination above 2° (checked every physics substep). With attachment flags b_i, completed-lift flag ℓ, final position t_T, yaw error $\delta\theta_T$ and minimum measured assembly clearance $d_{\min}$, mission success is

$$S = \mathbf{1}\{b_1 b_2 = 1,\ \ell = 1,\ \|t_T - t_g\| < 0.10,\ |\delta\theta_T| < 0.10,\ d_{\min} \geq 0,\ \neg\mathsf{F}\}. \tag{18}$$

Units are meters and radians. Acquisition failures and rejected routes remain in the denominator. Planned clearance (20 mm) and measured execution clearance (0 mm) are distinct thresholds.

B. Grasp space and matched search

The archived 50 mm perimeter enumeration screens 2,110,211 opposed layouts across nine scenes; 45,917 (2.18%) pass the reduced geometric constraints (Fig. 4a). Only 0.53–4.32% survive within individual scenes. This sparsity makes carrier placement a design variable rather than an arbitrary initial condition. Of 54 selected perimeter layouts submitted to planning, 51 yield verified routes. A distinct end-face study confirms 95/108 sampled placements by whole-body IK and verifies 52/54 selected routes. These samples test successive filters; their denominators are not interchangeable or estimates over a continuous grasp space.

To isolate aperture sampling, we run 16 matched seeds with gate probability 0.35 or zero, holding geometry, steering, goal bias, margin and stopping budget fixed. Every returned path is independently verified at 5 mm swept resolution, including the tree connection. Aperture-biased search verifies **16/16** paths versus **5/16** for uniform search at 20,000 iterations (Fig. 4b). This single-scene result supports the sampling mechanism under the declared budget, not general search optimality. It is separate from the archived full-mission cohorts.

C. Geometric compatibility

The recorded beam configuration admits a 25.8° object-only yaw interval but only 5.5° for the assembly (Fig. 5). This evaluates the geometric consequence of including the carriers, not the relative efficiency of search algorithms.

A separate paired development comparison completes 36/40 missions with the original local disc-cover model and 40/40 after adopting the global rectangle geometry and reducing the local margin from 30 to 20 mm. The same actor and starts are used in both conditions; four living-

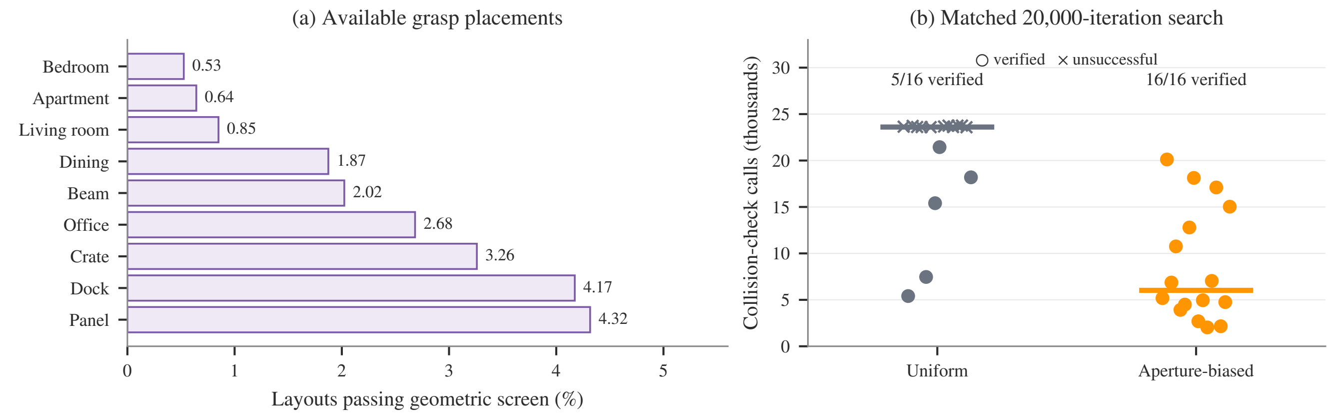

Fig. 4. Grasp geometry and controlled search. (a) Fraction of sampled opposed placements passing the geometric screen in each scene (45,917/2,110,211 total). (b) All 16 matched seeds per condition on the apartment scene. Circles are verified routes, crosses unsuccessful searches; bars show median check counts. Both conditions use 20,000 iterations, checked tree joins, 10 mm swept tests and independent 5 mm verification. A collision-check call may test a segment, so its count is not an iteration count or a wall-time measurement.

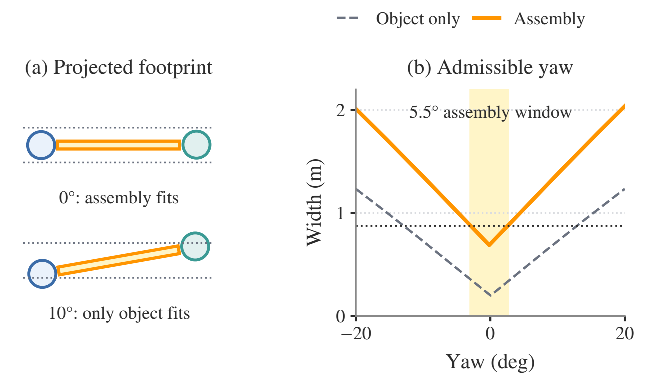

Fig. 5. Beam passage at the measured handoff configuration (seed 142024). Including the base footprints reduces the admissible yaw interval from 25.8° to 5.5°. Dashed bounds include the 22 mm search margin.

room stalls are resolved. Because geometry, margin and implementation settings change together, this is evidence of model incompatibility rather than a controlled estimate of either modification's effect. Protocols and traces are retained in the supplement.

D. Mission completion and failure localization

The frozen system completes **35/40 missions (87.5%)**. Of 40 starts, 37 achieve joint attachment, 36 complete lift, and 35 obtain and execute a verified route (Fig. 6). Failures comprise two panel tip-overs before grasp, one delayed beam acquisition timeout, one incomplete office-table lift and one exhausted living-room search. The conditional 35/35 route-execution result excludes upstream failures and is not the full-mission success rate. Four trials per scene provide descriptive counts, not precise scene-level reliability estimates.

With a 7.68 s activation gap in a loading-dock trial, attachment occurs at 1.88/10.76 s and transport starts at 13.44 s (supplementary acquisition trace). The supervisor permits limited early lift while the second robot approaches. A same-start panel replay succeeds under the geometric teacher where the frozen actor tips the object; this single diagnostic does not estimate overall teacher performance.

A second frozen cohort extends the scene set to twelve and completes **46/48** missions. It uses a different actor and upper/lower panel handles; both failures are acquisition timeouts. Together, the cohorts show where the integrated task succeeds and where it fails, but the 46/48 score is not a paired estimate of improvement over 35/40. All twelve scene counts and controller identities are retained in the supplement.

E. Clearance and computation

For the 31 successful doorway missions, median planned and measured minimum clearances are 32.5 and 29.0 mm; the median paired difference is 2.2 mm. The smallest measured clearance is 19.95 mm, above the execution threshold but below the planning margin. Across the 35 successful missions, the median per-trial maximum carrier drift is 16.2 mm and the overall maximum is 40.8 mm. Consequently, the nominal 20 mm margin cannot yield a uniform certificate through Eq. (4). These are control-step payload/base measurements, not continuous articulated-body clearance.

Among successful missions, median lift completion and first goal arrival are 10.6 and 95.4 s; transport occupies a median 88.6% of simulated duration. Median search-plus-smoothing wall time is 26.45 s (11.41–70.70 s), excluding verification and retries. Simulation pauses during planning, so this time is excluded from mission duration. Complete MPC latency and deadline-miss distributions are unavailable; configured update rates do not establish real-time operation.

F. Spatial tracking and physical validity

Separate probes use a 1.5 kg, $1.90 \times 0.09 \times 1.25$ m panel with supervised acquisition and welded retention. A 0.15 m clearance lift precedes 45°/90° roll commands lasting 22.5/45 s. The geometric teacher reaches 44.29°/88.29°. Completion requires 3 s of holding within 25 mm and 3°, linear/angular speeds below 0.04 m/s and 0.04 rad/s, retained contacts and no physical termination.

A separate spatial imitation actor completes 0° and 15° probes but terminates its 45° probe on self-penetration at

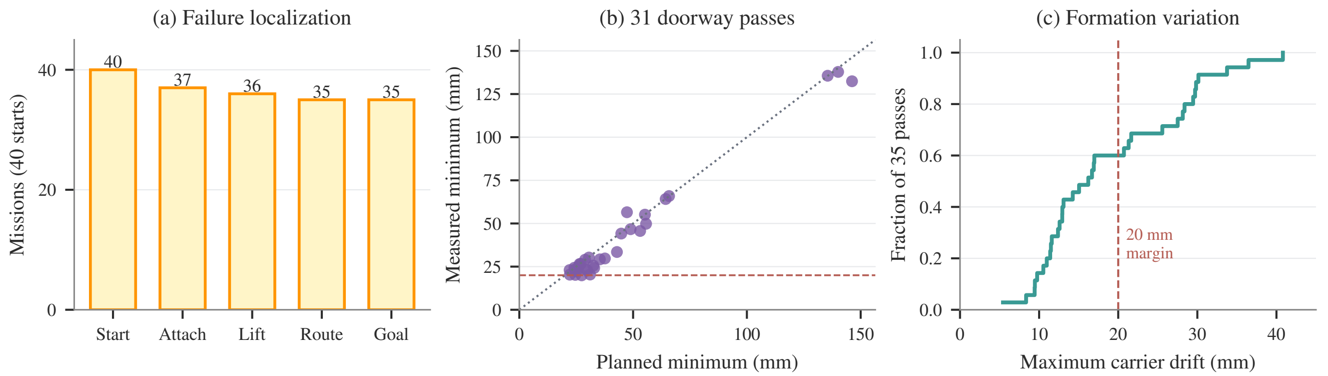

Fig. 6. Mission evidence at three levels. (a) All 40 starts remain in the stage counts. (b) Paired planned/measured minima for the 31 successful doorway missions; dotted diagonal indicates equality, dashed line the 20 mm planning margin. (c) Distribution of maximum carrier drift in all 35 successful missions. The four arena passes are included in (c) but have no finite doorway clearance.

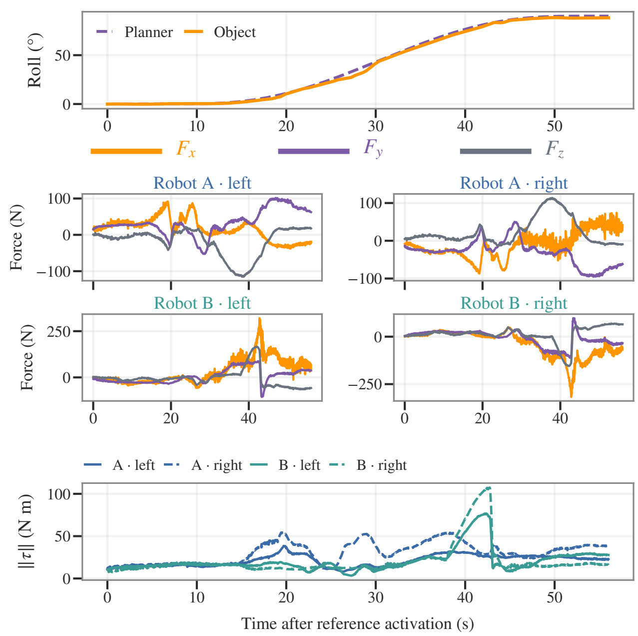

Fig. 7. Reference, measured roll and individual weld reactions in an instrumented 90° teacher probe. Forces use fixed lifted-object axes; moments are palm-centered. These are solver reactions, not force commands.

17.26°. Teacher and actor starts and handle spans differ, preventing a matched controller comparison. These selected positive-roll probes do not establish reliability throughout the prescribed ±90° envelope or demonstrate large-angle rotation during doorway passage.

An instrumented repeat reconstructs each hand's reaction from its weld Jacobian and multipliers (supplement). Despite reaching 88.29°, one hand reaches 363.2 N and another 107.4 N m (Fig. 7). Summed hand forces agree with logged robot totals within 10^{-6} N, but opposing reactions are obscured by aggregate feedback. Successful pose tracking therefore does not demonstrate feasible grip loading.

G. Native-contact lift learning

Calibration exposes a narrow usable contact region (Fig. 8a). At the selected reference, an idle policy passes 0/10 hold-pose probes and a scripted closure/proportional-lift controller passes 20/20. These establish that the criterion is attainable and requires action. They use different seeds from the learned evaluation and are not a matched ranking of controller reliability.

The 81,920-step PPO checkpoint passes **200/200** deterministic trials from a separate seed block, with a conditional 95% Wilson interval of 98.1–100% and median completion time 5.07 s. All four hands satisfy the pose/contact conjunction; Fig. 8c shows the terminal position errors rather than only the binary outcome. The same run's noisy training rollouts have much lower success (Fig. 8b), so training-window rates are not substituted for frozen-policy evaluation.

These are near-grasp, nominal-randomization-level lift results. They establish neither full-domain randomization performance nor recovery from distant starts. The checkpoint and evaluation-domain audit are supplied in the supplement. Earlier full-approach failures use different resets and hold criteria, so they are not a matched ablation of calibration or curriculum. Frictional-retention navigation and large-angle learned reorientation remain outside the demonstrated native-contact capability.

VII. Discussion and Conclusion

The experiments support a common interpretation: the grasp determines both the shape that must be planned and the physical target that local control must realize. The sparse placement space and matched sampling study show why grasp geometry belongs in planning. The mission cohorts show that acquisition and supervised progression belong in the evaluation denominator. The weld-load and native-reference studies show that accurate object motion alone does not establish a physically usable grasp.

The shared object reference connects these levels without making them identical. Carrier offsets are fixed planning parameters but vary during execution; the measured 40.8 mm maximum drift exceeds the nominal margin. Welded missions demonstrate supervised transport, whereas native-contact learning currently demonstrates near-grasp lift and hold. Neither shared weights nor centralized imitation removes the information conveyed by the supervisor. Matched teacher

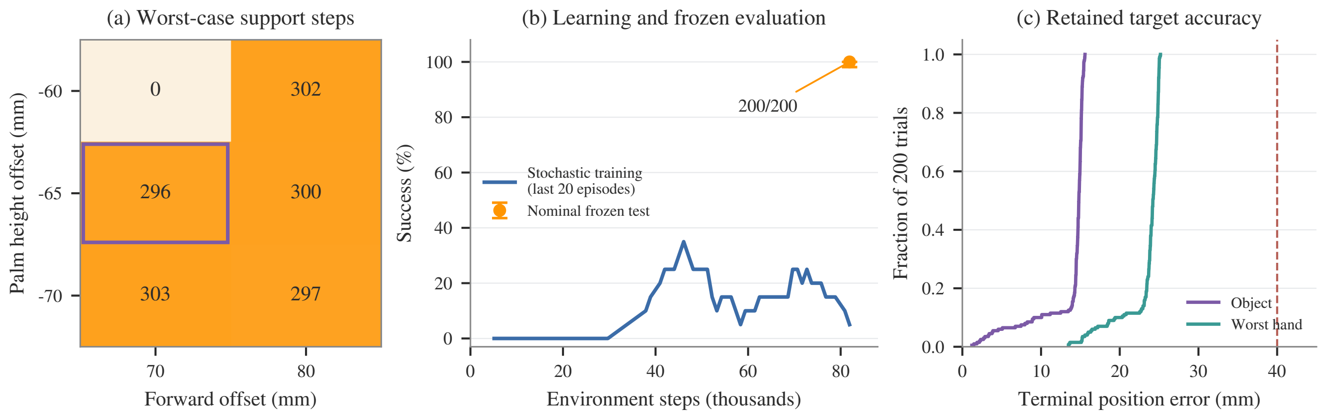

Fig. 8. Native-contact lift study. (a) Minimum simultaneous four-hand support steps out of 350 across three seeds and three masses per cell, at zero approach standoff; the outlined cell is the selected reference. (b) Stochastic training (rolling 20-episode mean) and the separate nominal-domain frozen test at 81,920 steps; the error bar is the test's Wilson interval. (c) Terminal object and worst-hand position errors in all 200 trials; the dashed line is the 40 mm acceptance threshold. The learned task starts near the grasp and caps base translation at 1 mm/s.

baselines, repeated training seeds and message ablations are still needed to attribute benefits to learning or coordination.

The next integration must preserve the calibrated grasp under approach, base motion and changing support, while planning and control run without pausing the physics. Full articulated clearance, actuator-limited loading, unseen layouts and hardware execution require separate validation. Within its stated simulation scope, M2 provides a connected formulation and failure-inclusive evidence from grasp placement through cooperative motion, with a contact-based learning stage that makes the remaining physical gap explicit and measurable.

References

[1] D. Alvear, G. Turkiyyah, and S. Park, "Cooperative grasping for collective object transport in constrained environments," arXiv:2509.03638, 2025.

[2] H. Zhang, H. Song, W. Liu, X. Sheng, Z. Xiong, and X. Zhu, "A novel semi-coupled hierarchical motion planning framework for cooperative transportation of multiple mobile manipulators," *Robotica*, vol. 43, pp. 4302–4324, 2025.

[3] B. Pandit *et al.*, "Learning Decentralized Multi-Biped Control for Payload Transport," in *Proc. Conf. Robot Learn. (CoRL)*, 2025, pp. 1021–1034.

[4] B. Pandit, A. K. Shrestha, and A. Fern, "Multi-Quadruped Cooperative Object Transport: Learning Decentralized Pinch-Lift-Move," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2026.

[5] J. Kim, R. T. Fawcett, V. R. Kamidi, A. D. Ames, and K. Akbari Hamed, "Layered Control for Cooperative Locomotion of Two Quadrupedal Robots: Centralized and Distributed Approaches," *IEEE Trans. Robot.*, vol. 39, no. 6, pp. 4728–4748, 2023.

[6] R. Doshi, H. R. Walke, O. Mees, S. Dasari, and S. Levine, "Scaling Cross-Embodied Learning: One Policy for Manipulation, Navigation, Locomotion and Aviation," in *Proc. Conf. Robot Learn. (CoRL)*, 2025, pp. 496–512.

[7] R. Doshi, T. Gao, A. Chen, C. Finn, and J. Bohg, "CHORUS: Decentralized Multi-Embodiment Collaboration with One VLA Policy," *arXiv:2606.12352*, 2026.

[8] Y. Chen, R. Qiu, Z. Li, Y. Zhou, G. Ren, and W. Zhi, "Robots that collaborate: Sequential asymmetric imitation for learning coupled robot policies," *arXiv:2606.16490*, 2026.

[9] T. An, F. De Vincenti, Y. Ma, M. Hutter, and S. Coros, "Collaborative Loco-Manipulation for Pick-and-Place Tasks with Dynamic Reward Curriculum," *arXiv:2509.13239*, 2025.

[10] K. Kedia, T. G. W. Lum, J. Bohg, and C. K. Liu, "SimToolReal: An Object-Centric Policy for Zero-Shot Dexterous Tool Manipulation," *arXiv:2602.16863*, 2026.

[11] T. G. W. Lum, K. Kedia, C. K. Liu, and J. Bohg, "Play2Perfect: What Matters in Dexterous Play Pretraining for Precise Assembly?" *arXiv:2606.26428*, 2026.

[12] D. P. Losey, M. Li, J. Bohg, and D. Sadigh, "Learning from My Partner's Actions: Roles in Decentralized Robot Teams," in *Proc. Conf. Robot Learn. (CoRL)*, 2020, pp. 752–765.

[13] Y. Liu, R. Leib, W. Dudley, A. Shafti, A. A. Faisal, and D. W. Franklin, "The role of haptic communication in dyadic collaborative object manipulation tasks," *arXiv:2203.01287*, 2022.

[14] Z. Cao, T. An, C. Li, S. Coros, and M. Hutter, "PAINT: Partner-Agnostic Intent-Aware Cooperative Transport with Legged Robots," *arXiv:2604.12852v1*, 2026.

[15] S. M. LaValle, *Planning Algorithms*. Cambridge University Press, 2006.

[16] J. J. Kuffner and S. M. LaValle, "RRT-Connect: An efficient approach to single-query path planning," in *IEEE Int. Conf. Robotics and Automation (ICRA)*, 2000, pp. 995–1001.

[17] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Nob Hill Publishing, 2017.

[18] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: An operator splitting solver for quadratic programs," *Math. Program. Comput.*, vol. 12, no. 4, pp. 637–672, 2020.

[19] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.

[20] MuJoCo Contributors, "MuJoCo 3.5.0 documentation: Equality constraints," https://mujoco.readthedocs.io/en/3.5.0/computation/index.html#equality, 2026, accessed Sep. 6, 2026.

[21] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv:1707.06347*, 2017.

[22] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proc. Int. Conf. Artif. Intell. Stat. (AISTATS)*, ser. Proceedings of Machine Learning Research, vol. 15, 2011, pp. 627–635.